Elizabeth Hall, PhD

Perception Scientist

I'm a PhD-trained Perception Scientist specializing in the rapid development and evaluation of XR hardware and sensing systems for perception enhancement. I have developed novel technologies for 6DoF eye, body, and object tracking in VR/AR through the integration of geometric computer vision, optical imaging, and contextual sensing. I also have experience with RF-based solutions, and generative audio rendering for enhanced spatial awareness. I am committed to delivering next-generation sensing and perceptual systems that remain genuinely human-centered.

Contact

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Expertise

multi-sensory 6DoF for AR/VR
3D geometry
localization
eyetracking
device and display calibration
object detection and tracking
body motion tracking / simulation
semantic segmentation
image processing / distortion
machine learning
Kalman filters
camera projection
holography
LLMs
vision-language models
neural networks
transformer models
LSTM / forecasting / RNN
AI - mediated reality
Unity / Unreal
electroacoustics

Experience

May 2025 - Jan 2026
Meta Reality Labs Research via Inspyr
Hardware UX Researcher

- Led 7 targeted studies quantifying speech enhanced Voice Quality on hardware prototypes; shared recommendations with leaders and enabled the on-time launch of Conversation Focus on Meta Ray-Bans at Connect 2025
- Defined KPI protocols and evaluated e2e latency metrics for real-time communication on Meta AI glasses over Bluetooth enabled broadcast
- Spearheaded the development and Meta-wide distribution of a VR/AR research platform for assessing immersive telepresence with lifelike, generative multisensory avatars
- Managed a team of 4 assistants and interns to evaluate advanced spatial audio rendering and ray-tracing algorithms on integrated and distributed AR/VR systems with limited compute resources
- Designed evaluative experiments for all stages of prototype-to-production development; experience working with tactile transducers, digital sensors, and the Martini sync protocol for device pairing over Bluetooth

Spatial Audio for AR and VR

Conversation Focus with Meta Ray Ban Glasses

Sept 2018 - Dec 2024
UC Davis Integrated Attention Lab
Graduate Research Fellow

- Led a 4-year research program on multisensory attention in VR, securing over $230K in funding from the *Office of Naval Research*
- Managed and trained 10 assistants on eyetracking and display calibration
- Developed a psychophysical framework for assessing 3D depth perception and image distortion
- Designed 6DoF object detection and eye-tracking systems analysis across multiple VR and AR branded devices
- Created and evaluated vision-language transformer models of 30k semantic segmentations across images of complex, cluttered environments
- Published 7 scientific papers in high-impact journals and invited to present by *ICCV*, *Psychonomic Society*, and the *Vision Science Society*

New VR Research on Attention

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Programming

Python
MATLAB
R
SQL
C++
Javascript
HTML / CSS
Linux
NumPy
SciPy
PyTorch
Tensorflow / Keras / CUDA
OpenCV
OpenGL
OpenCL / parallel processing
Docker
AWS
S3
ETL

Education

2018 - 2024
PhD Perception Science
UC Davis

2015 - 2016
MSc Cognitive Neuroscience
University of the Basque Country

2011 - 2015
BA Psychobiology
Bennington College

Certifications

AFNI for fMRI

via Neuromatch Academy:
Computational Neuroscience (2020)
Deep Learning (2021)
NeuroAI (2024)

first aid / CPR
wilderness first aid
swiftwater rescue

Experience

May 2023 - Sept 2023
Amazon Alexa Customer Behavior
Data Science Intern

- Developed an LSTM to predict device engagement and behavior from 10+ million Alexa users with new releases across Entertainment, Shopping, and Smart Home
- Reweighted features to align forecasts with A/B test results, achieving 93% accuracy
- Built feature engineering pipeline to process millions of customers' data through Amazon data-storage system improving data processing for Amazon Devices weekly telemetry dashboard

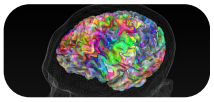
Sept 2019 - Sept 2023
Defense Advanced Research Projects Agency (DARPA)
Statistical Consultant

- Statistical consulting work in partnership with the University of Melbourne for the DARPA SCORE and Replicats project
- Supported the development and deployment of automated tools that enable DoW personnel to quickly calibrate the validity of behavioral science claims / results

Nature News: Automated tools to rate research responsibly

Sept 2016 - June 2018
National Institutes of Health
Research Fellow

- Designed online experiments to collect and evaluate human data and semantic segmentations using JavaScript and Amazon Mechanical Turk, collecting data from over 2,000 participants
- Recorded 7T BOLD fMRI signal and applied multivariate pattern analyses and neural network based-decoding to identify stimulus perception within the visual cortex and hippocampus
- Published neural computing and neuroimaging research in *Nature Communications*, *Scientific American*, *Cerebral Cortex* and *Quanta Magazine*



Quanta Magazine: New Map of Meaning in the Brain Changes Ideas About Memory

Scientific American: Our Memory Is Even Better Than Experts Thought

August 2015 - August 2016
BCBL - Basque Center on Cognition, Brain and Languages
Graduate Student Researcher

- Applied beamforming and time-frequency analysis to magnetoencephalographic signal; characterized the auditory steady-state response in children with Dravet Syndrome epilepsy